Gauge refinement: exact steps, scale dependence and the infrared obligations

Maintained scope. Series convolution is exact; transverse elimination produces an effective interaction that must be retained and controlled. Refining a lattice adds ultraviolet modes. A physical mass gap and confinement are separate infrared claims: respectively a spectral bound and an area law. The estimates below organize the proof dependencies; no four-dimensional continuum gap is proved by this map.

One-loop transmutation supplies a possible physical scale, not a positive spectral coefficient. Normalization, volume uniformity, generated interactions, continuum reconstruction and visibility to physical observables remain explicit obligations. A coarse pushforward does not determine its fine conditional laws. Statuses refer to the linked written arguments or identified literature; internal model reviews are not independent peer review or novelty evidence.

1. Exact operations and arbitrary cuts

For spacing a_μ in D Euclidean dimensions, use heat times $t_{\mu\nu} = \lambda_D a_\mu a_\nu / \prod_{\rho\neq\mu,\nu} a_\rho$ in the small-field normalization of the series/parallel theorem. Halving one direction produces:

- **Series:** a face containing that direction splits into two half heat times. Integrating its new edge convolves weights; character coefficients multiply. Heat kernels close exactly in every dimension.
- **Parallel:** each transverse face acquires a copy at doubled heat time. The mid-plane integral is a $(D-1)$-dimensional gauge factor Ψ with coarse-field-conditioned bridge edges. Pointwise products add logarithms, but heat-kernel closure has compact image defects.
- **Count:** there are $\binom{D-1}{2}$ parallel planes per directional halving. An isotropic step has D directional moves and engineering heat time $t = \lambda_D a^{4-D}$, before running is included.

A cut at any $s \in (0,1)$ convolves heat times st and $(1-s)t$. Trapezoid dual lengths keep transverse copies at heat time $2t$; the bridge variance is $s(1-s)t_e$, and the free parallel defect is multiplied by $4s(1-s)$. The series/parallel note, Corollaries 2_s and 5_s, gives the exact free calculation and its $U(1)$ extension. A cut schedule's physical scale change includes both daughters and their dual volumes; assigning $\log(1/s)$ to one daughter alone loses that information.

For compact $U(1)$ the bridge is a positive winding mixture; weights are s-independent and shifts $2\pi(1-s)W$ give $\mathbb{Z}_q$ at $s = p/q$. For simply connected compact groups the exact character formula is a signed Weyl-polynomial Cartan-image expansion, not a proved positive mixture of homotopy sectors. At a rational cut p/q, the reached subgroup of the $SU(N)$ centre is $\mathbb{Z}_{\gcd(q,N)}$. Thus dyadic cuts see the $SU(2)$ sign but only the identity in the $SU(3)$ centre; trisection sees triality. A central mid-edge flip creates thin flux on incident faces. Whether those image terms organize physical centre vortices remains open.

Newton's insertion is the series analogue. One cut spends $3s(1-s)K_\tau$, $K_\tau = F^2\tau^3/(24m)$; complete mesh refinement exhausts the cell correction. Its cut geometry proves the additive shares and conditional Gaussian record information. This analogy imports no gauge gap.

2. Dimension and group map

Dimension	Exact part	Retained correction or scale	Surviving conclusion and proof status
$0+0$	One compact integral	$e^{-c/t}$ image structures	Gross–Witten large-N transition is a different limit; dimension ladder
$1+0$ mechanics	Constant-force series law after cubic scalar correction	$-F^2uv(u+v)/(8m)$ insertion defect	Supplied-$\hbar$ propagator, or classical motion at zero; refinement
$1+1$ gauge	Heat-kernel series convolution; no parallel plane	General actions retain a conjugation-invariant Lévy exponent $\psi(R)$	Circle spectrum $\hbar cL\psi(R)$; Yang–Mills under the specified Lindeberg condition; no local particle gap; zero-spacing theorem
$1+2$ gauge	One parallel plane; exact free Schur defect	Relative free defect $a^2K^2/16+(Ga)^2/8$; compact $U(1)$ image density $e^{-\pi^2/(8t)}$	At fixed physical λ_3, compact $U(1)$ tends to a free photon; $SU(N)$ gap $C_3\hbar c\lambda_3$ remains open
$1+3$ gauge	Three parallel planes; exact series move	Engineering-marginal $t=\hbar g_{\mathrm{cl}}^2$ runs logarithmically; Λ is the transmutation scale	Nontrivial continuum theory and $C\hbar c\Lambda>0$ remain open; four-dimensional composition
$D>4$	Series convolution still exact	$t=\lambda_D a^{4-D}$ grows on refinement	No ultraviolet small-t regime by this scaling; higher-dimensional lattice phases are not a constructed continuum theory

The detailed $1+2$ distinctions are:

Group	One-cell or one-step control	Remaining dependency
Free $\mathbb{R}^n$	Exact Gaussian elimination and Schur composition	This blocking's full continuum comparison must be specified
$U(1)$	Exact charge form and vortex bound; normalized whole-measure TV bound $2(L/a)^3e^{-\pi^2/(6\lambda_3 a)}$	Continuum trajectory matters; Villain/monopole proof
$SU(2)$	Midpoint characters at every spin; exact spin-1/2 and spin-1 cube sectors with image bounds	Full spin sum and physical recoupling; exact midpoint

Group	One-cell or one-step control	Remaining dependency
$SU(2)$ mid-plane	Formal order-t shifts: cut positive, transverse negative; normalized barriered weak-strip estimates	Nonlinear source estimate, recoupling, contact cancellation and iteration; small-field proof, order-t expansion
$SU(3)$	Group-general local curvature term $t_j\|X_j\|^2/512$; formal bulk coefficients 3/2 times $SU(2)$	Large fields, exact physical matching, stable iteration and a positive infrared coefficient

The circle rotor and a zero-dimensional integral

The exact $1+1$ circle spectrum $E_R = \hbar c\lambda_2 L C_2(R)/2$ equals $\hbar^2 C_2(R)/(2I)$ with $I = 1/(cg_{\rm cl}^2 L)$, an inertia independent of $\hbar$ at fixed classical coupling. The infinite-line theory instead has no local particle; its static-charge string tension is $\hbar c\lambda_2 C_2/2$. The zero-spacing theorem fixes the conventions.

A $0+0$ integral has no Hamiltonian time evolution. Its large-N transition is a limit in matrix size, not a continuum particle gap. For a stable quartic integral, asymptotic perturbation theory alone does not establish failure of Borel summation; the earlier unqualified claim of non-Borel-summability is withdrawn. No such claim is needed for the refinement argument.

3. Why ultraviolet refinement bears on an infrared gap

Let $\xi = \hbar/(mc)$ be a putative physical correlation length and L the box size. A continuum limit sends $a/\xi \to 0$ at fixed ξ/L, followed by a separate infinite-volume limit when required. The lattice gap is $\delta = aE/(\hbar c)$; for $E = mc^2$ it is a/ξ. A physical string tension has a different dimensionless combination $a^2\sigma/(\hbar c)$. A halving halves the former and quarters the latter if the physical quantities survive. It does not calculate their positive coefficients.

For $D = 4$, write $t = \hbar g_{\rm cl}^2$ and $b_0 = 11N/(48\pi^2)$. At one loop,

$$\mu\frac{dt}{d\mu} = -2b_0 t^2 + O(t^3), \qquad \Lambda = \mu e^{-1/(2b_0 t(\mu))}.$$

The displayed Λ relation uses the one-loop truncation; higher orders add prefactors and scheme dependence. A measured coupling fixes one normalization of Λ. A scheme change rescales it and the numbers C_i inversely so that a putative $E_i = C_i\hbar c\Lambda$ is invariant. Neither running nor dimensional analysis proves $C_i > 0$.

Exponential corrections and the spectral coefficient

Within the one-loop relation,

$$e^{-\chi/t} = (a\Lambda)^{2b_0\chi}.$$

A dimension-one lattice gap would have exponent $2b_0\chi = 1$. An image term with exponent greater than four can be suppressed even after counting $O((L/a)^4)$ cells, provided its normalized boundary-uniform bound is proved. Root-image exponent $\chi = 8\pi^2$ gives $11N/3$; the $SU(3)$ trisection centre exponent $24\pi^2$ gives 33. The dislocation criterion $\chi > 2/b_0$ is 6/11 of the instanton action for $SU(2)$. These powers are a formal scaling diagnostic for the stated image exponents, not uniform estimates for every large field. The zero-spacing treatment and centre calculation preserve their precise scopes.

Exponentially small terms have vanishing Taylor coefficients at zero coupling; a perturbative power series cannot establish their positive spectral weight. The earlier analogy to renormalon

positions was sourced only to metadata and memory and supplies no identification of these image terms with a renormalon. It is not used as evidence.

Below four dimensions $\lambda_D = \hbar g_{\mathrm{cl}}^2$ carries length power $D-4$, and dimensional energy is $\hbar c \lambda_D^{1/(4-D)}$. This scales as $\hbar^{3/2}$ in $1+1$ and $\hbar^2$ in $1+2$ at fixed classical coupling. At $D=4$ the running equation supplies an exponential scale instead of this dimensional power. This is an RG statement, not a derivation by formally setting $D=4$ in the divergent exponent. The rotor and quartic matrix model have their own $\hbar^2$ and $\hbar^{4/3}$ laws, proved in the low-dimensional comparison.

Two uniformities and a constructed measure

The number of steps between a and a fixed ξ grows as $\log_2(\xi/a)$. Refinement adds weak-coupling steps at the ultraviolet end. A fixed range of physical scales near ξ still requires infrared control. The confinement-scale bands collect the finite-lattice Wilson-action evidence; numerical bands or a finite verification proposal are not a certified mixing theorem.

Uniformity in refinement depth as $a \to 0$ and uniformity in box size as $L \to \infty$ are separate. If p_* is the coarse pushforward, then $\langle F \rangle_{p_* \mu} = \langle F \circ p \rangle_\mu$ exactly, but many fine conditional measures have that same coarse law. A compatible refinement family must be constructed. Transport of a coarse-loop channel also requires spectral visibility to reach a gap for all relevant physical observables. Continuum nontriviality and reconstruction remain further steps.

4. Supplied action, lattice gaps and physical limits

Compact $U(1)$ in $1+2$ has a lattice monopole gap that vanishes in the fixed-λ_3 continuum limit (Gross). Along the Göpfert–Mack scaling $\lambda_3 a \simeq c_0/[2\log(1/a)]$, $c_0 \simeq 4.99$, a massive Gaussian limit instead survives with divergent physical monopole density. The monopole note gives the source and trajectory qualifications. A lattice gap alone is thus insufficient.

Physical $\hbar$ is supplied in both Gaussian theories; it is not an independently derived lower bound on every action difference. Removing a mass gap need not remove quantum kinematics. Conversely a generated scale need not imply a mass gap: massless QCD under spontaneous chiral symmetry breaking has pions, and an infrared fixed point has gapless infrared behavior. The two-dimensional delta potential has a transmuted binding energy and a continuum above zero; its bound-state energy is not a field-theory mass-gap proof. Low-dimensional models and the action/gap comparison retain these distinct mechanisms. Confinement and a physical spectral gap each require their own argument.

5. Remaining proof obligations

- **Nonabelian spatial blocking:** extend the retained-barrier estimates (112), (116) and centered quadratic source test (177) to the full nonlinear source bound (113), physical recoupling, contact cancellation, target-covering paths and perturbed-action iteration. The small-field notebook holds the constants and proofs; its individual stages are not independent confirmations of the target theorem.
- **Four-dimensional depth:** control generated cubic/quartic kernels, subtracted contractions, endpoint matching and the accumulated remainder. The conditional composition theorem gives the averaged $-2b_0 \log 2$ coefficient only under those hypotheses. One-step parallel matching and Schur composition prove different parts.
- **Time-only halving:** at fixed spatial lattice, electric heat kernels compose in series and exponential magnetic weights add in parallel. Their noncommutation is controlled in the strong Trotter limit for the compact-group Laplacian plus bounded magnetic potential (Trotter 1959; Chernoff 1968; Kogut–Susskind 1975 and Creutz 1977, metadata in the

source companions). A norm rate needs separate commutator domains. Spatial continuum uniformity does not follow from this fixed-lattice Hamiltonian construction.

- **Cut schedules:** free-field shape convergence and nonabelian generated weights require control beyond the one-cut factor $4s(1-s)$. Whether triadic $SU(3)$ refinement helps organize large fields is open.
- **Records in the mechanical analogy:** Gaussian marks and complete controllers retain conditional noise, conjugate recoil and weighted covariance budgets. A total precision sum alone does not characterize arbitrary changing triangular arrays. The score model passes its branch-dependent composition test at every supplied coefficient, including zero; composition alone selects no positive action.
- **Infrared and reconstruction:** a nontrivial physical continuum measure, volume-uniform spectral positivity, observable visibility and an independent area-law argument remain necessary. Mass-gap position and continuum obligations state the targets.

6. Consequence for STATE

Exact refinement identities locate the missing estimates; they do not replace them. Keep ultraviolet control, infrared spectral positivity, confinement, construction of the continuum theory and the supplied action unit separate. The linked arguments hold the proofs and source evidence; this map supplies only their dependencies and scaling conventions.